Marmara University

ALPEREN TURAKOĞLU 150719663
ENES SEZER 150719541
ENES ŞENGÜL 150719714
ENSAR YILMAZ 150719655
ERKAN ERTEN 150719654

MEASUREMENT OF DISSIPATION FACTOR AND DIELECTRIC CONSTANT OF SOLID DIELECTRICS

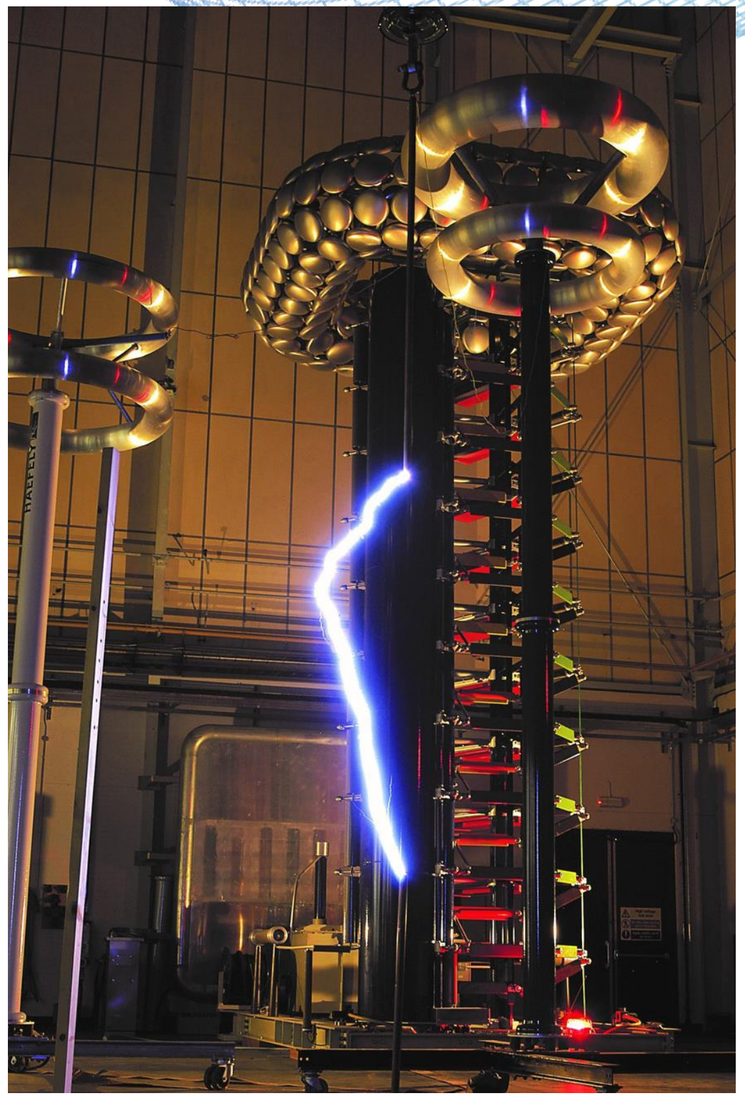
THE PURPOSE OF MEASUREMENT OF DISSIPATION FACTOR AND DIELECTRIC CONSTANT OF SOLID DIELECTRICS.

- **It increases the efficiency of dielectric materials in high voltage applications.**
- **Material selection and design must comply with safety and performance standards.**
- **The dissipation factor and dielectric constant are used to evaluate the quality and applicability of the dielectric material.**



THE PURPOSE OF MEASUREMENT OF DISSIPATION FACTOR AND DIELECTRIC CONSTANT OF SOLID DIELECTRICS.

- **This test is used to calculate the dispersion factor and dielectric constant of solid dielectric materials.**
- **These factors are critical in evaluating the electrical properties and performance of dielectrics.**
- **It examines the quality of the material and its insulating properties.**



DIELECTRIC DISSIPATION FACTOR MEASUREMENT OF POWER EQUIPMENT UNDER DISTORTED EXCITATION VOLTAGE

Insulation assessment is crucial for the health of high voltage power system equipment. The dielectric response (DR) of dielectric materials to applied voltage is indicative of insulation condition. The dielectric dissipation factor (DDF) measures real power loss and aids in diagnosing insulation condition. Field service knowledge informs testing standards for assessing insulation in various equipment types.



DIELECTRIC DISSIPATION FACTOR MEASUREMENT OF POWER EQUIPMENT UNDER DISTORTED EXCITATION VOLTAGE

While traditional methods consider pure sinusoidal voltage excitation, this study introduces a framework for measuring DDF under nonsinusoidal excitation. A circuit model accounts for polarization loss, and mathematical formulas quantify the impact of harmonics on DDF measurement. Laboratory testing and simulations validate the results, focusing on 33kV dry-type epoxy resin cast current transformers.



THE EQUIPMENTS OF THE TEST



AMPLIFIER



OSCILLISCOPE



SIGNAL GENERATOR



MULTIMETER

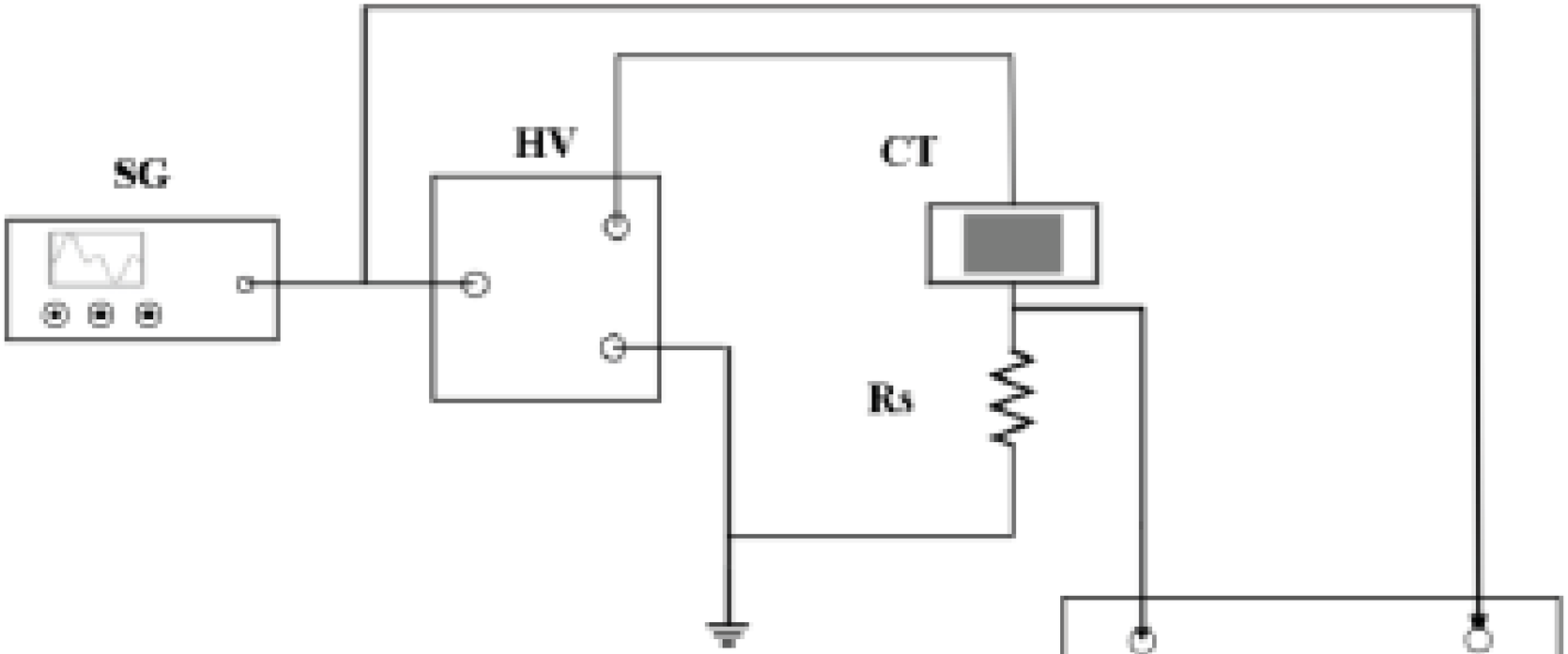
THE EQUIPMENTS OF THE TEST



SHUNT



TWO 33KV DRY TYPE EPOXY RESIN CAST CURRENT TRANSFORMERS (CT-1 AND CT-2)



SG	Signal Generator
HV	High Voltage Amplifier
CT	Current Transformer
Rs	Shunt
DSO	Digital Storage Oscilloscope

DDF MEASUREMENT CIRCUIT DIAGRAM

TEST STEPS

1

Preparing the Measuring Devices

- **An LCR (inductance, capacitance, and resistance) measuring device**
- **A source (usually an oscillator)**
- **A measuring probe**

2

Sample Preparation & Positioning

- **The surfaces of the sample must be clean and smooth**
- **the dimensions and geometry of the sample must match the requirements of the measuring instrument**

3

Dissipation Factor Measurement

- **The Dissipation factor is** calculated by measuring the phase loss **of the signal from the sample.**

4

Dielectric Constant Measurement

- **The dielectric constant is a quantity related to the dielectric property of the sample and is calculated** based on the capacitive response **of the signal from the sample.**

5

Analysis of Results

- **Values are used to understand the dielectric performance.**
- **Values can be compared to evaluate the material's insulating properties, energy storage capacity, and performance in other applications.**

FORMULAS

$$\tan \delta = \frac{P_{diss}}{P_{in}}$$

P_{diss} *is the power lost in the dielectric in watts.*

P_{in} *is the input power applied to the dielectric in watts*

Dissipation Factor Calculation

$$\varepsilon = \frac{C}{\varepsilon_0 A}$$

The symbol ε *represents the dielectric constant*

C *is the dielectric capacitance in farads*

ε_0 *is the epsilon note constant and has a value of approximately* $8.854 \times 10^{-12}\ F/m$.

A *is the area of the electrodes on the dielectric in* m^2

Dielectric Constant Measurement

EXAMPLE

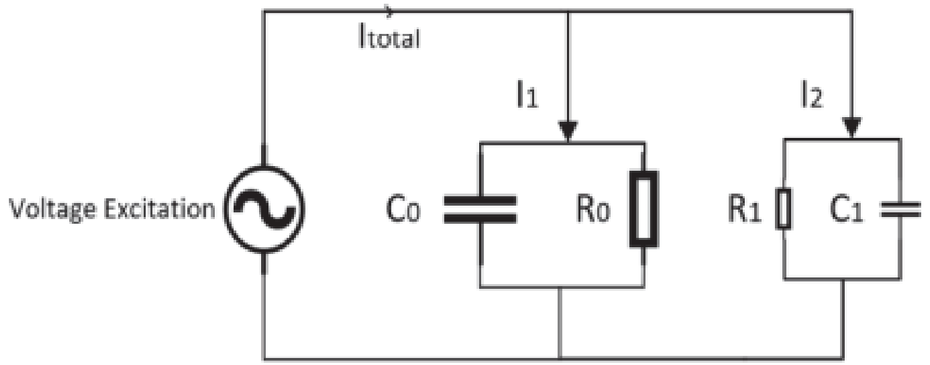


Figure 1. Leaky capacitor model with parallel polarization branch

$$DDF = \frac{P}{Q} \longrightarrow DDF = \text{The dielectric dissipation factor} \quad P = Real\ Power \quad Q = Reactive\ Power$$

$$Y = j\omega C_{eq} + \frac{1}{R_{eq}} = j\omega C_0 \left(\varepsilon_r^{/} - j\varepsilon_r^{//}\right)$$

$$R_{eq} = \frac{R_0 R}{R_0 + R} \quad \text{and} \quad C_{eq} = C_0 + C$$

$$DDF = \tan\delta = \frac{I_{R_{eq}}}{I_{C_{eq}}} = \frac{1}{\omega R_{eq} C_{eq}}$$

$$DDF = k \quad \frac{1}{\omega R_{eq} C_{eq}}$$

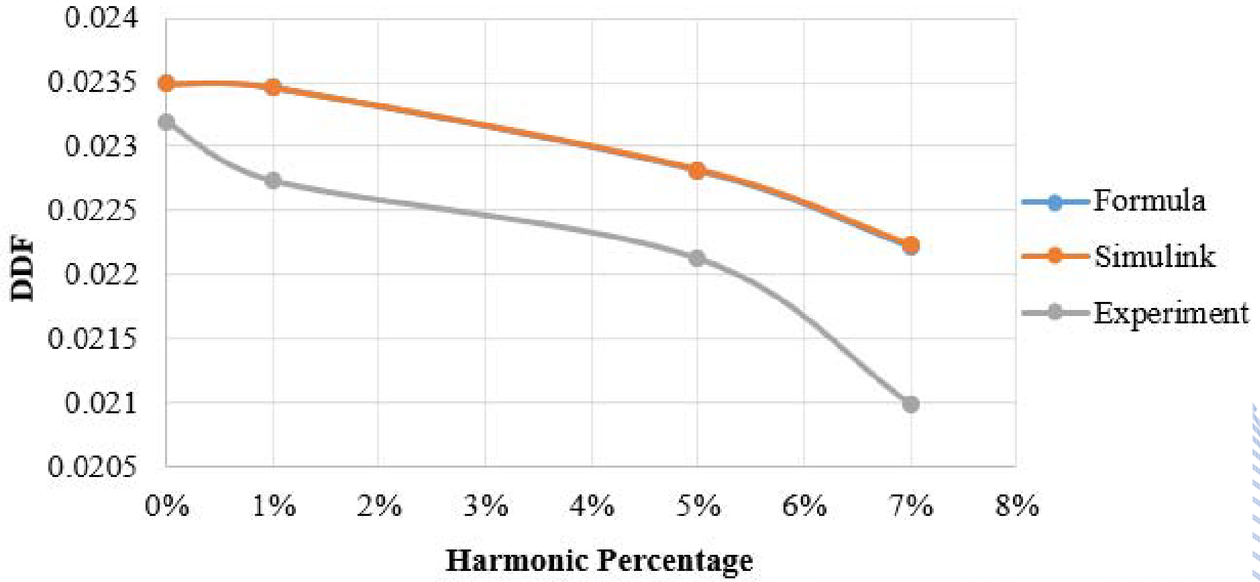
WHAT ARE THE RESULTS AND FINDINGS OF THE TEST

These test results aid in selecting, optimizing, and designing solid dielectric materials for electrical and electronic applications. They provide insights into performance characteristics, energy losses, and energy storage capabilities of dielectrics, enabling the development of efficient and reliable electrical systems.

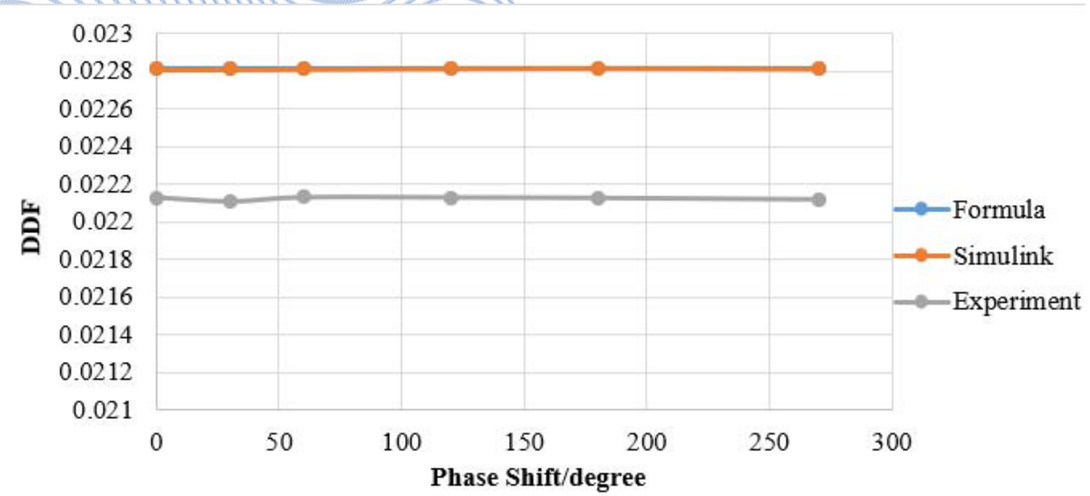


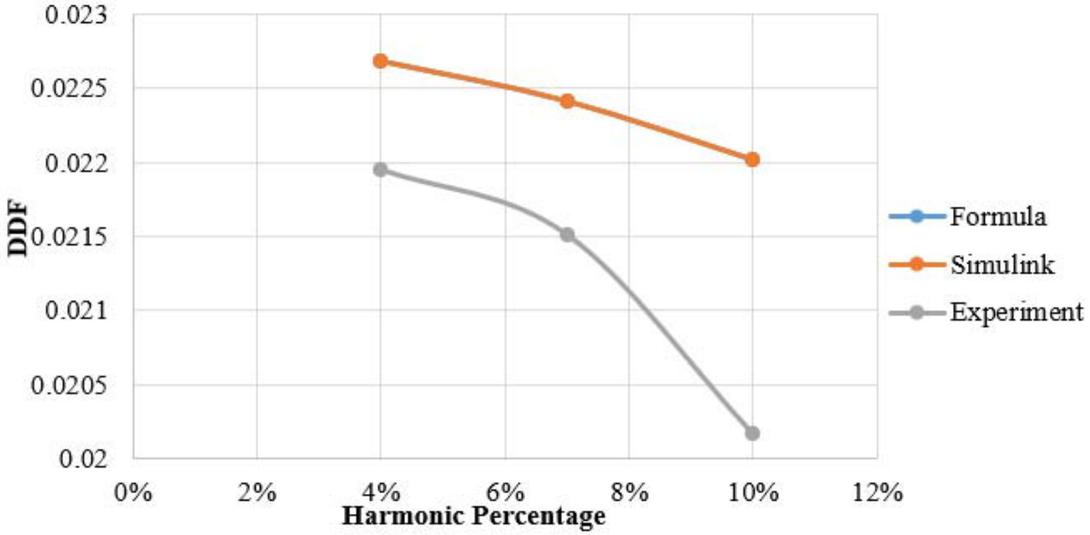
WHAT ARE THE RESULTS AND FINDINGS OF THE TEST FOR ARTICLE

In a previous study, the dissipation factor (DDF) decreased with increasing harmonic levels in an artificial test object. However, the experimental results differed from analytical calculations, suggesting that the polarization property in real high voltage equipment is more complex and challenging the assumption of constant equivalent capacitance with frequency in high harmonic scenarios



WHAT ARE THE RESULTS AND FINDINGS OF THE TEST FOR ARTICLE





The DDF shows a consistent decrease as the proportion of harmonics increases, with a fixed 5% of the 5th harmonic and varying quantities of the 3rd harmonic. However, at higher harmonic percentages (10%), the rate of change deviates noticeably from the expected mathematical formula, as seen in the Figure's termination at 7% but still displaying similar variations.

WHAT ARE THE RESULTS AND FINDINGS OF THE TEST FOR ARTICLE

The study investigates polarization loss and DDF under sinusoidal and distorted excitation voltages. It finds a decrease in DDF with increasing proportion of harmonic components. Further research is needed to explore the specific polarization effects caused by harmonics in high voltage equipment.

STANDARDS FOR TEST:

ASTM D150

IEC 60270

IEEE Std 1434

IEC 60093

UNIVERSITY OF MANCHESTER, UK
High Voltage Laboratory

- Established in 1950, the High Voltage Research Laboratory at the University of Manchester is the largest high voltage laboratory in the UK.
- With its capacity to test equipment for 400Kv power systems, it continuously upgrades its facilities to stay up-to-date with evolving technologies.
- Serving small businesses, large industry organizations, and collaborating with government, the lab focuses its research on providing real-life solutions to various challenges.



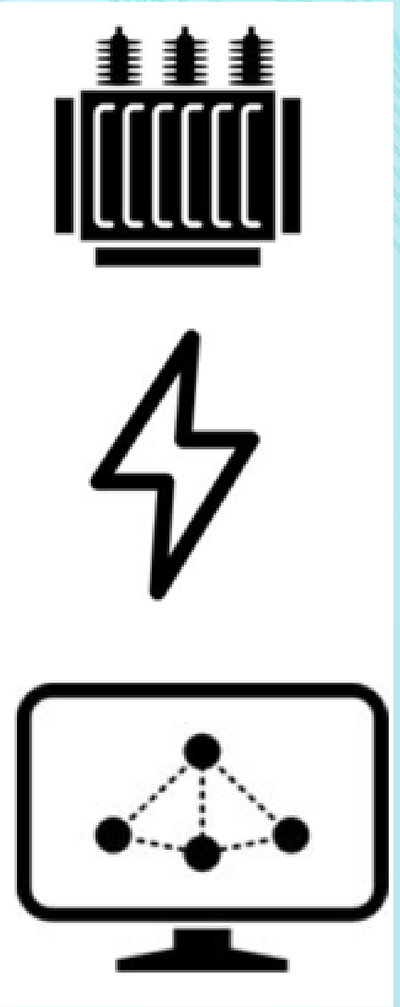


THERE ARE MANY KINDS OF DIFFERENT EQUIPMENT IN THIS LABORATORY. AMONG THEM, THE MAIN PRODUCTS ARE AS FOLLOWS;

- 2MV impulse generator
- 800kV AC test set
- 600kV DC test set
- 20kVA high current source (configurable to maximum current of 10kA)
- Salt fog and environmental test chambers
- Modern digital measurement equipment
- Material processing and characterisation equipment
- Radio Influence Voltage (VIR) measurement systems.
- UV inspection facilities for corona monitoring
- High voltage environment chambers for testing of equipment in harsh environments
- Anechoic chamber, with 200kV input for noise and corona measurements.
- Software simulation capabilities (FEA modelling, PSCAD, CDEGS)

SOME OF THE AREAS BEING WORKED ON ARE;

- **Transformers ; they have developed new methods to study how these assets age and anticipate their lifespan.**
- **Novel insulation; To resist the higher voltages required for the massive power transmission, this requires new and improved insulators. In terms of creating and evaluating these new materials, they are at the forefront.**
- **Condition monitoring; Real-time monitoring allows us to determine whether we can push these devices further in an emergency or predict when they will fail. To do this, they are creating the solutions.**



WHICH STANDARDS ARE FOLLOWED DURING THE TESTS?

IEC 60060:

IEC 60270:

IEC 60364:

IEC 60502:

IEC 60694:

IEC 61083:

IEC 62271:

ARE THERE ANY SOFTWARES OR DIGITAL TOOLS USED IN THE LAB? WHAT ARE THEY AND FOR WHICH ACTIVITIES THEY ARE USED FOR?

Simulation and analysis software such as PSCAD, EMTP-RV, and ATP-EMTP enable researchers and engineers to simulate and analyze electrical systems, predict high voltage system behavior, evaluate parameter influence, and optimize designs.

PSCAD

EMTP-RV

The reference for power systems transients

Measurement and Data Acquisition Systems: During testing, voltage, current, and other electrical parameters are captured and recorded using digital instruments including oscilloscopes, power analyzers, and data loggers. These instruments aid in taking accurate measurements and evaluating how equipment performs in various scenarios.

Partial Discharge (PD) measuring Systems: To find and examine partial discharge activity in electrical equipment, PD measuring techniques and software are utilized. These systems aid in evaluating the insulation's state and locating any potential faults or weaknesses.

Software for Control and Monitoring: Software for Control and Monitoring is frequently used in high voltage test equipment. These software programs include functions like automated testing, real-time monitoring, data logging, and remote control.

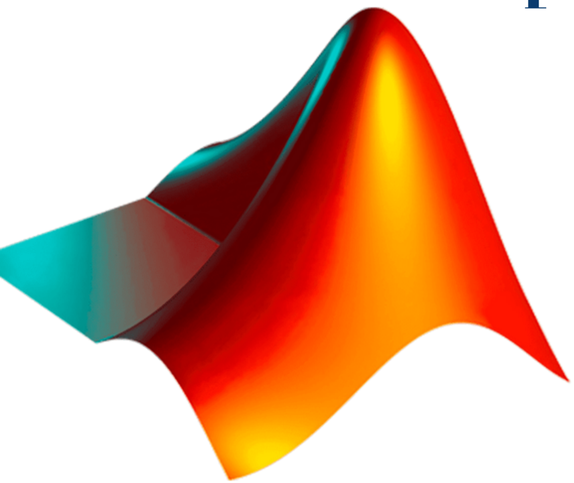
- **Computer-Aided Design (CAD) Software: High voltage components, systems, and testing sets are designed and modeled using CAD programs like AutoCAD or SolidWorks. They support visualizing and optimizing equipment architecture and configuration.**





AUTOCAD

- **Analysis and Reporting Software: Software for data analysis and reporting, such as MATLAB, Python, or Excel, is frequently used for processing statistical data and producing reports based on experimental findings.**



MATLAB®



python™

THANKS FOR YOUR PATIENCE

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